

**ASSIGNMENT No: 19**

TITLE: Develop a Java application to establish connectivity with a database and perform data retrieval using the SELECT operation.

Problem Statement

Develop a Java application to establish connectivity with a database and perform data retrieval using the SELECT operation.

Learning Objectives

To establish database connectivity and retrieve records using JDBC SELECT statements.

Theory

Java Database Connectivity (JDBC) is a standard Java API that enables communication between Java applications and relational databases. It provides a platform-independent mechanism to connect to databases such as MySQL, Oracle, PostgreSQL, and SQL Server. Before executing SQL statements, a JDBC driver is loaded and a connection is established using the DriverManager class. The Connection object represents an active connection to the database, while the Statement object is used to execute SQL queries. The **SELECT** statement retrieves records from database tables based on specified conditions. The retrieved records are stored in a ResultSet object, which allows sequential access to the data. JDBC simplifies database programming and is widely used in desktop, web, and enterprise applications.

Code:

```
package E19;

import java.lang.*;
import java.sql.*;

public class E19 {

    static String url = "jdbc:mysql://localhost:3306/symbiosis";
    static String user = "root";
    static String password = "12345";

    public static void main(String[] args) {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
```

```

Connection con = DriverManager.getConnection(url, user, password);

Statement stmt = con.createStatement();

ResultSet rs = stmt.executeQuery("SELECT * FROM students");

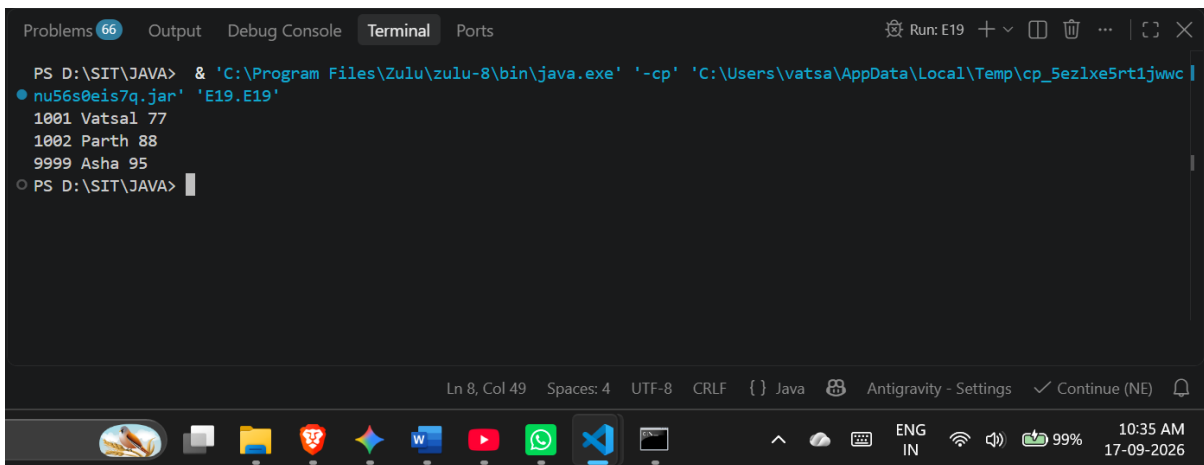
while (rs.next()) {
    System.out.println(
        rs.getInt("id") + " " +
        rs.getString("name") + " " +
        rs.getInt("marks")
    );
}

con.close();

} catch (Exception e) {
    System.out.println(e);
}
}
}

```

Output:



```

PS D:\SIT\JAVA> & 'C:\Program Files\Zulu\zulu-8\bin\java.exe' '-cp' 'C:\Users\vatsa\AppData\Local\Temp\cp_5ezlxe5rt1jwwc |
nu56s0eis7q.jar' 'E19.E19'
1001 Vatsal 77
1002 Parth 88
9999 Asha 95
PS D:\SIT\JAVA>

```

Exercise

1. Create a Java application to display **student records from database** using SELECT query

Code:

```
package E19;

import java.lang.*;
import java.sql.*;

public class StudentRecords {

    static String url = "jdbc:mysql://localhost:3306/SIT_JAVA";
    static String user = "root";
    static String password = "12345";

    public static void main(String[] args) {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            Statement stmt = con.createStatement();

            ResultSet rs = stmt.executeQuery("SELECT * FROM students");

            System.out.println("ID\tName\tMarks");
            System.out.println("-----");

            while (rs.next()) {
                System.out.println(
                    rs.getInt("id") + "\t" +
                    rs.getString("name") + "\t" +
                    rs.getInt("marks")
                );
            }

            con.close();

        } catch (Exception e) {
            System.out.println(e);
        }
    }
}
```

Output:

```
PS D:\SIT\JAVA> java -cp 'C:\Users\vatsa\AppData\Local\Temp\cp_Se2lxe5rt1jwwc\nu56s0eis7q.jar' 'E19.StudentRecords'
```

ID	Name	Marks
1	Aarav	85
2	Priya	92
3	Rohan	78

```
PS D:\SIT\JAVA>
```

2. Create a Java application that connects to a database and displays **product details** using a **SELECT query**. Show product ID, product name, quantity, and price.

Code:

```
package E19;

import java.lang.*;
import java.sql.*;

public class ProductDetails {

    static String url = "jdbc:mysql://localhost:3306/SIT_JAVA";
    static String user = "root";
    static String password = "12345";

    public static void main(String[] args) {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(url, user, password);
            Statement stmt = con.createStatement();

            ResultSet rs = stmt.executeQuery("SELECT * FROM products");

            System.out.println("ID\tProduct Name\t\tQuantity\tPrice");
            System.out.println("-----");

            while (rs.next()) {
                int id = rs.getInt("product_id");
```

```

String name = rs.getString("product_name");
int qty = rs.getInt("quantity");
double price = rs.getDouble("price");

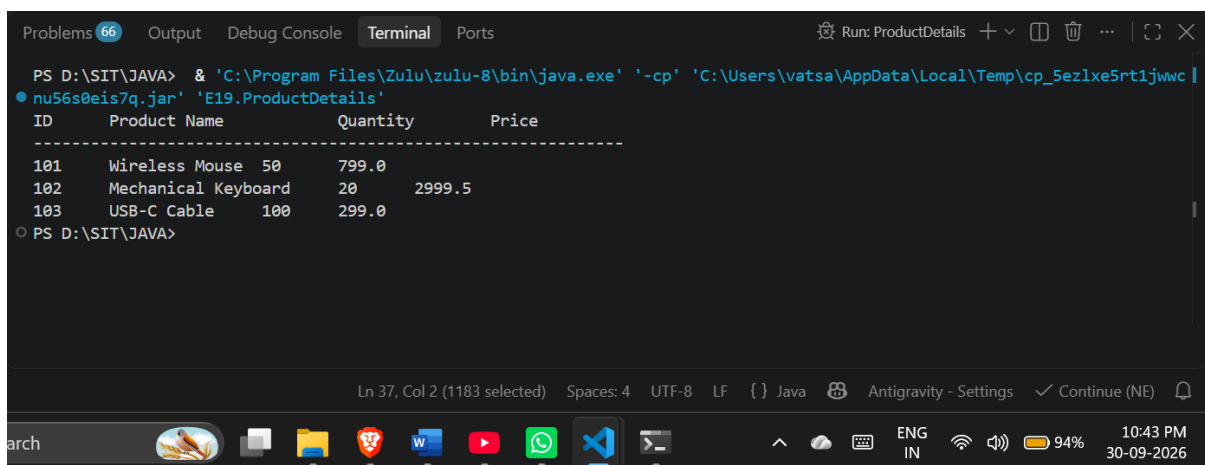
System.out.println(id + "\t" + name + "\t" + qty + "\t" + price);
}

con.close();

} catch (Exception e) {
    System.out.println(e);
}
}
}

```

Output:



```

PS D:\SIT\JAVA> & 'C:\Program Files\Zulu\zulu-8\bin\java.exe' '-cp' 'C:\Users\vatsa\AppData\Local\Temp\cp_Se2lxe5rt1jwwc |
nu56s0eis7q.jar' 'E19.ProductDetails'
ID      Product Name      Quantity      Price
-----
101     Wireless Mouse    50           799.0
102     Mechanical Keyboard 20          2999.5
103     USB-C Cable       100          299.0
PS D:\SIT\JAVA>

```

Conclusion:

I learned how to successfully connect a Java application to a MySQL database using the JDBC API by configuring the MySQL connector driver in my project workspace. I learned how to establish a database connection using `DriverManager.getConnection()` and use the `Statement` interface to execute `SELECT` queries. Additionally, I learned how to iterate through a `ResultSet` to extract specific column data using methods like `getInt()` and `getString()`, and how to properly format and print these database records to the console.